Heat

I] Textbook Questions:

⭐ 1 Mark Objectives: ⭐

1. Fill in the blanks and rewrite the sentence.

a. The amount of water vapour in air is determined in terms of its **absolute humidity**.
b. If objects of equal masses are given equal heat, their final temperature will be different. This is due to difference in their **specific heat**
c. During transformation of liquid phase to solid phase, the latent heat is **released**.

⭐ 2 Marks Questions: ⭐

Q. Explain:

1. In cold regions in winter, the rocks crack due to anomalous expansion of water.

Ans:

1. When water cools from 4 ^{0}C to 0 ^{0}C, it expands due to the anomalous behaviour of water.
2. So, in winter, when water inside the pipes cools below 4 °C, it expands.
3. This causes increase in pressure inside the pipes, and hence the pipes burst.
4. Similarly, when the water trapped in the cracks in a rock gets cooled below 4 °C, it expands and pushes the crack further apart. As a result, the rock cracks.

2. While deciding the unit for heat, which temperatures interval is chosen? Why?

Ans:

1. The units of heat are Joule (J) in SI units, cal (calorie) in cgs units. The amount of heat necessary to raise temperature of **1 g** of water by **1 ^{0}C** from **14.5 ^{0}C to 15.5 ^{0}C** is called one cal heat.
2. Similarly, the amount of heat necessary to raise the temperature of **1 kg** of water by **1 ^{0}C** from **14.5 ^{0}C to 15.5 ^{0}C** is called one kcal heat. It is clear that (**1 kcal= 1000 cal**).
3. If we heat **1 kg** of water by **1^{0}C** in different temperature range than **14.5 ^{0}C to 15.5 ^{0}C**, the amount of heat required will be slightly different.
4. It is, therefore, essential to define a specific temperature range while defining the unit of heat. Calorie and Joule are related by following relation: **1 cal = 4.18 Joule**

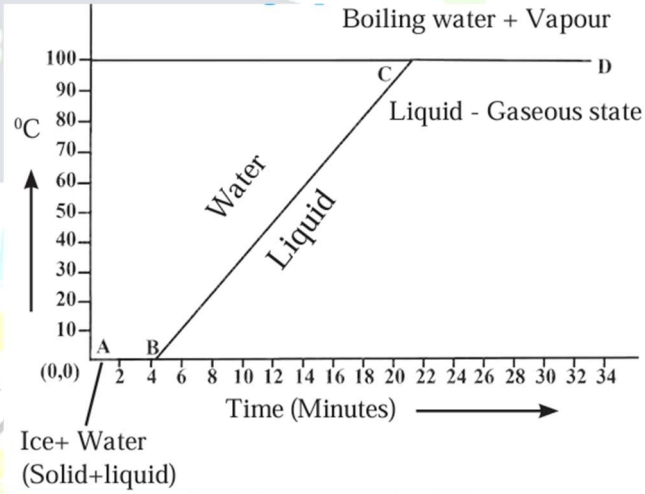
⭐ 3 Marks Questions: ⭐

Q. Answer the following.

1. Explain the following temperature versus time graph. [2019, 2022, 2024]

Ans:

1. **AB – Melting of ice:** Ice and water coexist at **0°C**. Heat is absorbed (Latent heat of fusion), but the temperature remains constant until all the ice melts.
2. **Melting point:** The constant temperature at which ice changes into water is called the **melting point of ice (0°C)**.
3. **Latent heat of fusion:** The heat absorbed at constant temperature during the conversion of solid into liquid is called **latent heat of fusion**. Per unit mass, it is called **specific latent heat of fusion**.
4. **BC – Heating of water:** After all the ice melts, the temperature of water increases from **0°C to 100°C** as heat is supplied.



5. **CD – Boiling of water:** At **100°C**, water changes into vapour at constant temperature. Heat is absorbed (Latent heat of vaporization) to convert the liquid into the gaseous state.
6. **Latent heat of vaporization:** The heat absorbed at constant temperature during conversion of liquid into gas is called **latent heat of vaporization**. Per unit mass, it is called **specific latent heat of vaporization**.

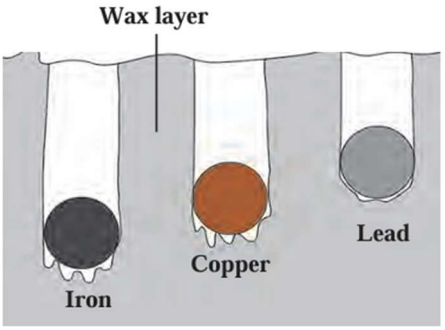
2. What is the role of anomalous behaviour of water in preserving aquatic life in regions of cold climate? [2024]

Ans:

1. Water has maximum density at **4°C**. Below **4°C**, water expands instead of contracting.
2. During winter, the surface water of lakes and ponds cools and reaches **4°C**, becoming denser and sinking.
3. As the surface water cools below **4°C**, it becomes less dense and remains at the top.
4. The surface water eventually freezes at **0°C**, forming a layer of ice.
5. Ice is a poor conductor of heat, so the ice layer acts as an insulating blanket and prevents the water below from freezing completely.
6. Thus, water remains liquid below the ice layer, allowing aquatic plants and animals to survive even in extremely cold climates.

3. **What is meant by specific heat capacity? How will you prove experimentally that different substances have different specific heat capacities?** [2025]

Ans.



5.8 Specific heat capacity of metals

1. Three spheres of **iron, copper and lead** having equal mass are taken.
2. The three spheres are placed in **boiling water** for some time, so that all reach **100°C**.
3. The hot spheres are immediately placed on a **thick layer of wax**.
4. The **iron sphere goes deepest** into the wax, the **copper sphere goes to an intermediate depth**, while the **lead sphere goes the least**.
5. This shows that for the **same mass and same rise in temperature**; the three metals absorb different amounts of heat. This property is called **specific heat capacity**.
6. **Specific heat capacity** is the amount of heat energy required to raise the temperature of **unit mass of a substance by 1°C**. Its SI unit is **J/kg °C**.

⭐ 5 Marks Questions: ⭐

1. **Read the following paragraph and answer the questions.** [2026, 2020]

If heat is exchanged between a hot and cold object, the temperature of the cold object goes on increasing due to gain of energy and the temperature of the hot object goes on decreasing due to loss of energy. The change in temperature continues till the temperatures of both the objects attain the same value. In this process, the cold object gains heat energy and the hot object loses heat energy. If the system of both the objects is isolated from the environment by keeping it inside a heat-resistant box (meaning that the energy exchange takes place between the two objects only), then no energy can flow from inside the box or come into the box.

i. Heat is transferred from where to where?
Ans: Heat is transferred from hot object to cold object.

ii. Which principle do we learn about from this process?
Ans: From this process we learn about the principle of heat exchange.

iii. How will you state the principle briefly?
Ans: Heat energy lost by the hot object = Heat energy gained by the cold object. This is called as 'Principle of heat exchange'.

iv. Which property of the substance is measured using this principle?

Ans: The specific heat of the substance is measured using this principle.

II] Extra - Previous Years' Questions: (PYQs)

⭐ 1 Mark Objectives: ⭐

MCQs:

1. The SI unit of heat is _____. [2024]
   (A) Calorie **(B) Joule** (C) Kcal/ kg ^{0}C (D) Cal/ g ^{0}C

2. **True or False**: The volume of water at 4 ^{0}C is maximum: **(False)** [2026]

3. **Match the pairs**: [2026]

Group A	Group B
Thermometer	a. Mechanical energy **b. To measure temperature** c. To measure electric current

4. **Name the behaviour of water between its temperature from 0 ^{0}C to 4 ^{0}C.** [2025]

Ans: The behaviour of water between its temperature from 0 ^{0}C to 4 ^{0}C is called anomalous behaviour of water.

5. **Write the use of Calorimeter.** [2024]

Ans: A calorimeter is used to measure the specific heat capacity of an object.

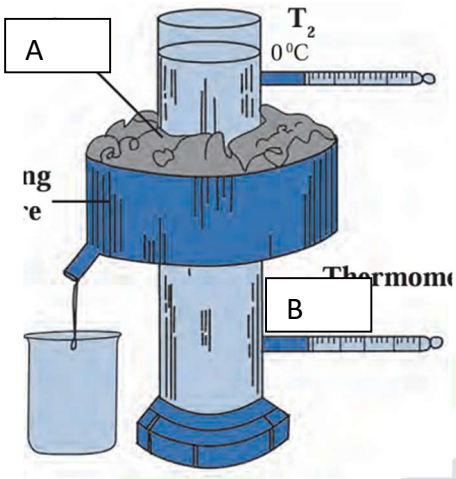
6. **What is meant by anomalous behaviour of water?** [2022, 2024]

Ans:

1. Water shows a special and exceptional behaviour. If we heat water from 0 ^{0}C up to 4 ^{0}C, it contracts instead of expanding.
2. At 4 ^{0}C its volume is minimum (due to contraction). If the water is heated further, it expands and its volume increases.
3. The behaviour of water between its temperature from 0 ^{0}C to 4 ^{0}C is called anomalous behaviour of water.

⭐ 2 Marks Questions: ⭐

1. **Label the following diagram and name the property that can be studied using it** [2026]



Ans:

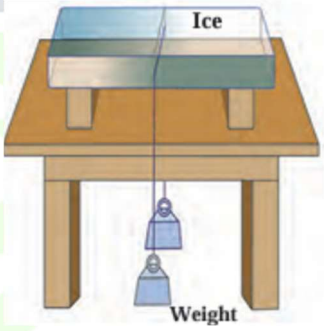
1. **A**: Freezing mixture **B**: 4 °C
2. Anomalous behaviour of water can be studied using it.

2. Define specific heat of an Object. Write its SI and CGS units. [2025]

Ans:

1. The amount of heat energy required to raise the temperature of a unit mass of an object by 1 °C is called the specific heat of that object.
2. **SI Unit**: J /kg °C **CGS Unit**: Cal /g °C

3. Observe the figure and answer the questions given below. [2024]



(a) Name the phenomenon shown in the figure

Ans: Regelation

(b) Define the above phenomenon. [2026]

Ans: The phenomenon in which the ice converts to liquid due to applied pressure and then re-converts to ice once the pressure is removed is called **regelation**.

4. How do we feel about air in each of the following conditions? [2023]

(a) Relative humidity is more than 60%.

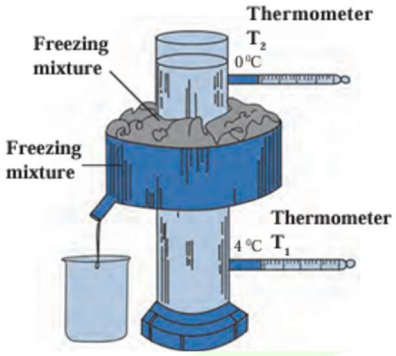
Ans: We feel that the air is humid.

(b) Relative humidity is less than 60%.

Ans: We feel that the air is dry.

5. Draw a neat labelled diagram of Hope's Apparatus. [2022]

Ans:



⭐ 3 Marks Questions: ⭐

1. **(a) What is meant by Regelation?** [2026]

Ans: The phenomenon in which the ice converts to liquid due to applied pressure and then re-converts to ice once the pressure is removed is called **regelation**.

(b) What is the effect of pressure on melting point of ice?

Ans: An increase in pressure lowers (decreases) the melting point of ice. Therefore, ice melts under pressure and refreezes when the pressure is removed.

(c) Write any one example of regelation.

Ans: eg. When two ice cubes are pressed hard together, they join to form a single piece of ice due to regelation.

⭐ 5 Marks Questions: ⭐

Observe the diagram and answer the following questions. [2023]

Refer the diagram of Hopes apparatus.

a. **What is the name of the given apparatus?**

Ans: The name of this is Hope's Apparatus.

b. **Which phenomenon is studied with the help of this apparatus?**

Ans: Anomalous behaviour of water is studied with the help of this apparatus.

c. **What are the final temperatures in the thermometers T_1 and T_2?**

Ans: The final temperatures in thermometers: T_1 is 4 ^{0}C and T_2 is 0 ^{0}C.

d. At what temperature the density of water is maximum?

Ans: The density of water is maximum at 4 ^{0}C.

e. Give one example of the above phenomenon in nature.

Ans: Survival of aquatic life in cold regions.

TO LEAD THE LEARNERS BEYOND THE STARS
MASTER CLASSES